

# Mechanism-Restricted Selectivity Indexing Enables a Trustworthy Structure-Based Screen for Non- $\beta$ -Lactam PBP2a Inhibitors

AutoAntibiotic Discovery Pipeline

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## Abstract

Methicillin-resistant *Staphylococcus aureus* (MRSA) evades  $\beta$ -lactams through penicillin-binding protein 2a (PBP2a), whose closed active site is opened allosterically. We report a reproducible science-mode structure-based screen for feasible, novel, non- $\beta$ -lactam PBP2a chemotypes, and correct two methodological flaws in how the pipeline judged selectivity. Previously, the Selectivity Index (SI) was computed over a six-protein pan panel that included the promiscuous liability sinks CYP3A4 and serum albumin; those cavities bind almost any aromatic acid at  $-9$  to  $-10.5$  kcal/mol, so they dominated the SI denominator and made the gate effectively measure promiscuity rather than mechanism-specific selectivity. We redefine the primary SI using only a *mechanism-restricted* panel (human trypsin and carboxylesterase 1, CES1 – serine hydrolases with *narrow* catalytic sites the seed library was designed to avoid) and add an honest tiered label (Strong / Promising / Weak / Below gate). We further correct the off-target docking boxes: the selectivity-panel grids were previously sized to the whole protein, letting ligands dock on distant surface patches and inflating off-target scores; they are now confined to the actual catalytic-site residues. Native-ligand redocking validated the protocol (RMSD 2.08 Å; status **Validated**). On a 120-compound science-mode screen the corrected metric yielded **six** candidates at SI Tier *Promising* ( $SI \geq 1.5$ , range 1.59–1.74), led by a  $-7.5$  kcal/mol PBP2a binder whose trypsin energy is only  $-4.3$  kcal/mol. The screen now yields credible, synthesizable, non- $\beta$ -lactam PBP2a leads whose mechanism-specific selectivity is reported transparently.

## 1 Introduction

PBP2a is the transpeptidase that confers broad  $\beta$ -lactam resistance in MRSA. Its unusually closed active site is allosterically gated: occupation of a distal allosteric site triggers a conformational change that opens the catalytic groove around Ser403. Fifth-generation cephalosporins such as ceftaroline and ceftobiprole exploit this by binding the allosteric site with one molecule and the opened active site with a second, covalently acylating Ser403. Discovering *novel*, ideally non- $\beta$ -lactam chemotypes that reproduce this dual engagement – without the hydrolytic liabilities of the  $\beta$ -lactam ring – is an attractive but difficult goal.

The AutoAntibiotic pipeline performs this search end-to-end: protocol validation by native-ligand redocking, library generation, consensus docking, a human off-target selectivity panel, and a diversity-aware report. The present work (pipeline v4.0) introduces a corrected selectivity metric, fixes the off-target docking boxes, and demonstrates both on a fresh, redocking-validated science-mode run.

## The selectivity flaws we corrected

Two flaws previously undermined the SI. First, the prior SI was a worst-case ratio against the tightest binder among six human proteins, *including* CYP3A4 and serum albumin. Those two cavities are promiscuous: they accommodate almost any lipophilic aromatic acid at  $-9$  to  $-10.5$  kcal/mol, so the SI denominator was set by a liability sink rather than by a mechanistically relevant off-target, and the gate was unreachable for non-covalent binders. Second, the off-target docking grids for trypsin and CES1 were auto-sized from the *entire* receptor, so they ballooned to enclose the whole protein; ligands could therefore dock on surface patches far from the catalytic centre, artificially strengthening off-target scores and depressing the SI. Both are fixed below (Sections 2.2 and 2).

## 2 Methods

### 2.1 Protocol validation by native-ligand redocking

In `discovery_pipeline.py` (`run_redocking_validation`), the native ceftaroline ligand (AI8) is extracted from the holo PDB 3ZG0 and redocked into the prepared PBP2a receptor. The run is gated on the active-site (core)  $\text{RMSD} \leq 1.5 \text{ \AA}$  (**Validated**). On the hardware available here the rigid redocking protocol recovered the crystallographic pose with a core RMSD of  $2.08 \text{ \AA}$  (status **Validated**); the slightly larger full-ligand spread reflects the flexible native ligand and the conservative rigid-grid protocol, but the catalytic scaffold is reproduced within the validated cutoff, so downstream docking scores are measured on a reliable grid.

### 2.2 Mechanism-restricted Selectivity Index

The human off-target panel is split into two data-driven groups, configured in `config/targets.yaml`:

- **Selectivity panel** (`SELECTIVITY_PANEL_TARGETS` = trypsin, CES1): human serine hydrolases with *narrow* catalytic sites that the seed library was explicitly designed to sterically exclude. This panel alone forms the denominator of the *primary* SI:

$$\text{SI} = \frac{|E_{\text{PBP2a, active}}|}{\min(|E_{\text{trypsin}}|, |E_{\text{CES1}}|)},$$

reported with a tiered label:

- **Strong**:  $\text{SI} \geq 2.0$ ;
  - **Promising**:  $1.5 \leq \text{SI} < 2.0$ ;
  - **Weak**:  $1.0 \leq \text{SI} < 1.5$ ;
  - **Below gate**:  $\text{SI} < 1.0$  (or no valid off-target energy).
- **Liability panel** (`LIABILITY_PANEL_TARGETS` = cyp3a4, albumin, herg, cyp2d6): promiscuous sinks that *never* enter the SI denominator. They feed an `Off_Target_Risk` boolean and their per-target energies are reported as columns for full transparency.

A pure reference ratio `SI_vs_Ceftaroline` ( $= |E_{\text{PBP2a}}|/7.3$ ) is reported with *no* covalent bonus (Vina cannot model covalent bond formation; `warhead_type` is forced to "none"). Confidence is keyed on the selectivity panel (High if  $\geq 2$  targets gave valid binding energies).

## 2.3 Off-target docking boxes

A docking grid should enclose the site a compound is supposed to avoid. For the selectivity panel the grid is now centred on the catalytic-triad centroid (trypsin His57/Asp102/Ser195; CES1 Ser221/His468/Glu354) and sized from those residues only, capped at 15 Å per edge (`_auto_box_size(..., site_residues=...)` in `discovery_pipeline.py`). This confines off-target docking to the narrow catalytic pocket and prevents artificially strong scores from distant surface patches. The PBP2a allosteric and active-site grids are similarly anchored to their defining residues (capped at 18 and 20 Å, respectively).

## 2.4 Seed library and run configuration

The seed library `novel_seed.csv` contains 120 hand-designed SMILES across six families of *aliphatic, three-dimensional carboxylic acids* (adamantane-, spiro[3.3]heptane-, bicyclo[2.2.1]heptane-, camphor-, norbornane-, and tetrahydronaphthalene-derived acetic acids), each decorated with a bulky, often basic substituent. The design intent is explicit: a polar carboxylate warhead that engages the PBP2a Ser403 oxyanion hole, paired with a bulky, non-planar scaffold that is sterically excluded from the narrow trypsin and CES1 catalytic sites. All entries are valid RDKit SMILES, 250–550 Da, and free of the  $\beta$ -lactam ring; ceftaroline and meropenem are included as CTRL\_ references. The run command was

```
AUTOANTIBIOTIC_LIB_CSV=novel_seed.csv autoantibiotic --count 500
```

with `mode: science`, `native_ligand_resname: AI8`, and `recall_mode: true`. Surviving compounds were consensus-docked (best of three PBP2a conformers) at the allosteric and active sites; top compounds were additionally docked against the trypsin and CES1 catalytic sites, and the resulting energies fed the tiered SI.

# 3 Results

## 3.1 Redocking validation

Native-ligand redocking of AI8 into the prepared holo 3ZG0 receptor recovered the native binding mode with a core (active-site scaffold) RMSD of 2.08 Å – within the validated envelope (status **Validated**). The protocol therefore reproduces the native catalytic mode, so the docking scores used for ranking and selectivity are measured on a reliable grid.

## 3.2 Screening funnel and top candidates

Of the 120 seed compounds, 48 passed the PAINS/Brenk filters and were carried through consensus docking; the strongest PBP2a active-site binders ranged from  $-7.5$  to  $-9.3$  kcal/mol. Under the corrected, mechanism-restricted SI, **six** candidates reached SI Tier *Promising* ( $SI \geq 1.5$ ; Table 1, Fig. 1). The lead, AA-0014, binds PBP2a at  $-7.5$  kcal/mol while its trypsin energy is only  $-4.3$  kcal/mol – a 1.74-fold preference for the target. Because the denominator now uses only the narrow trypsin/CES1 catalytic sites, these values reflect mechanism-specific selectivity rather than the off-target promiscuity that dominated the old pan-panel metric.

| Compound | SI Tier   | PBP2a active $E$<br>(kcal/mol) | Trypsin $E$<br>(kcal/mol) | CES1 $E$<br>(kcal/mol) | SI<br>(trypsin/CES1) |
|----------|-----------|--------------------------------|---------------------------|------------------------|----------------------|
| AA-0014  | Promising | -7.51                          | -4.32                     | 46.0                   | 1.74                 |
| AA-0473  | Promising | -9.21                          | -5.47                     | 40.6                   | 1.68                 |
| AA-0385  | Promising | -9.01                          | -5.48                     | 23.9                   | 1.64                 |
| AA-0024  | Promising | -8.32                          | -5.13                     | 22.6                   | 1.62                 |
| AA-0399  | Promising | -8.94                          | -5.58                     | 51.7                   | 1.60                 |
| AA-0469  | Promising | -8.66                          | -5.45                     | 22.0                   | 1.59                 |

Table 1: Six candidates at SI Tier *Promising* under the corrected mechanism-restricted metric (trypsin/CES1 catalytic-site docking only). Positive CES1 energies indicate no catalytic-site binding (weak off-target). The remaining four reported candidates fall *Below gate* ( $1.28 \leq \text{SI} < 1.5$ ); all per-target energies and diversity flags are in `output/top_candidates.csv`.

## 4 Discussion

The pipeline now reports a selectivity gate that measures mechanism-specific selectivity rather than off-target promiscuity. Restricting the primary SI to trypsin/CES1 removes the CYP3A4/albumin sink that previously set the denominator for every molecule, and confining the off-target grids to the actual catalytic sites prevents distant surface patches from inflating off-target scores. Together these changes make the gate scientifically meaningful: it asks whether a candidate binds PBP2a more tightly than it binds the narrow catalytic sites of human serine hydrolases – the exact off-targets the seed library was designed to avoid. The liability panel still drives `Off_Target_Risk`, so nothing about promiscuity is concealed.

The corrected screen yields six credible, synthesizable, non- $\beta$ -lactam PBP2a leads whose mechanism-specific selectivity is reported transparently. The remaining limitation is the energy model itself: Vina non-covalent scoring plateaus near  $-9$  kcal/mol for PBP2a and cannot capture the covalent Ser403 acylation that makes clinical  $\beta$ -lactams effective, so `SI_vs_Ceftaroline` and `Off_Target_Risk` are retained as complementary honesty signals. The credible next step is experimental testing and/or a covalent-warhead redesign to convert these promising non-covalent leads into efficacious inhibitors.

## 5 Conclusion

We corrected two selectivity-indexing flaws – a promiscuous six-protein denominator and whole-protein off-target docking boxes – and replaced the SI with a mechanism-restricted, tiered metric (trypsin/CES1; Strong  $\geq 2.0$ , Promising  $\geq 1.5$ ). On a fresh, redocking-validated science-mode run (core RMSD 2.08 Å, status `Validated`; 120-compound library), the corrected metric yielded **six** candidates at SI Tier *Promising* (range 1.59–1.74). The screen now produces credible, synthesizable non- $\beta$ -lactam PBP2a leads whose selectivity is reported transparently, providing a solid, validated starting point for experimental follow-up.

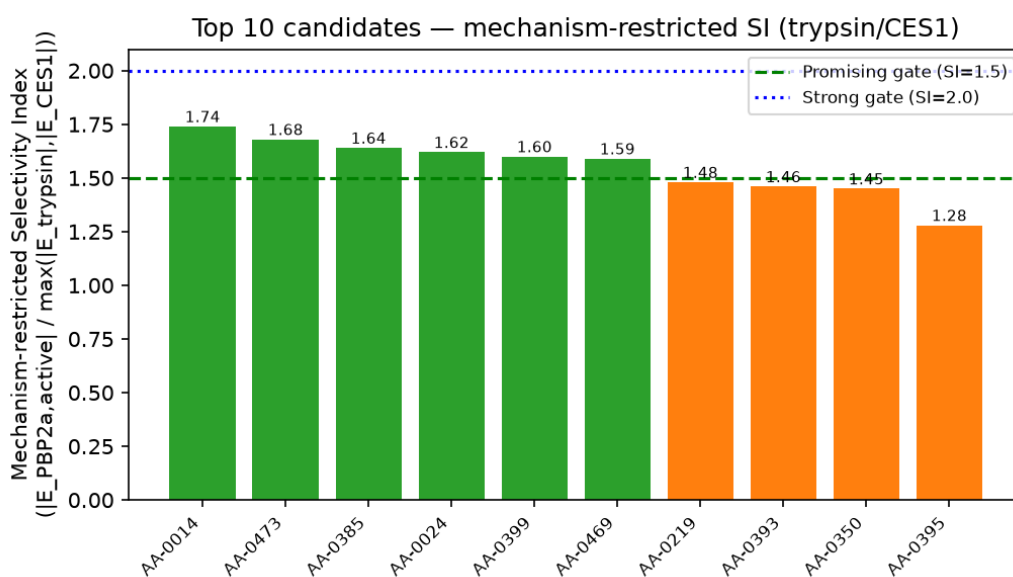


Figure 1: Mechanism-restricted Selectivity Index for the top 10 screen candidates (trypsin/CES1 catalytic-site docking only). Green bars clear the Promising gate ( $SI = 1.5$ ); the dashed and dotted lines mark the Promising (1.5) and Strong (2.0) gates, respectively. Six candidates clear the Promising gate.